Causal Forecast-Conditioned BS-RoFormer for Ocean Sound Separation

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Abstract

Ocean source separators normally infer a changing background only from the mixture being separated. We study whether a causal forecast of the expected background, computed from prior soundscape measurements, improves separation. The proposed system encodes historical one-third-octave levels, predicts the current background spectrum, and injects it into BS-RoFormer band tokens by concatenation, feature-wise linear modulation (FiLM), or cross-attention. We use time-disjoint deployment splits, paired SI-SDR evaluation, and a real-time factor check. A six-family surrogate study selects FiLM: it improves SI-SDR improvement (SI-SDRi) over an unconditioned separator by 0.34 dB on 768 paired examples, while JS divergence alone does not predict gain. We then use 42 hours of continuous real hydrophone audio and compare 5-s, 30-s, 1-min, and 10-min prediction windows with a fixed one-hour history. A validation rule retains persistence unless the learned forecast beats it. All resolutions yield significant paired separation gains on 1,024 held-out real-background clips; 5 s is best at +1.226 dB (95% interval $[1.082, 1.381]$), with real-time factor 0.0024. Only the 5-s learned forecast beats persistence on validation, showing that temporal conditioning helps even when elaborate dynamics do not.

1 Introduction

Passive acoustic backgrounds vary with shipping, weather, biology, and season. A separator that only observes the current mixture must estimate both the target and that background at once. Long-context soundscape forecasting offers an additional, strictly causal signal: what the background was expected to look like immediately before the mixture arrived. Our hypothesis is that this prior helps a separator reject predictable energy while retaining an unexpected injected source.

We combine the band-split RoPE Transformer (BS-RoFormer) architecture [4] with a causal recurrent forecast encoder. The contributions are: (i) a differentiable forecast–separation model; (ii) matched ablations of concatenation, FiLM, and cross-attention against no forecast; (iii) a leakage-audited deployment split; (iv) an evaluation protocol that reports separation, forecast error, uncertainty, and computational usability; (v) a six-family source generator with paired analysis by spectral shift, source–background overlap, and source entropy; (vi) a continuous real-audio study at four sub-hour prediction resolutions; and (vii) a reference-free, interventional evaluation on strongly annotated real whale calls.

2 Data and waveform construction

We use NOAA–Navy SanctSound hourly one-third-octave level (TOL) products [5]. MB03 deployments 01 and 02 form the training set, 03 is validation, and 04 is held-out test. A sample at hour t is eligible only when all H preceding hours and hour t are finite and exactly contiguous. Normalization statistics are fitted on the two training deployments only.

The public TOL product is an aggregate and cannot determine waveform phase. For each current spectrum S_t, we interpolate its dB envelope on FFT bins, draw a seeded random phase and Rayleigh-like magnitude jitter, invert the FFT, and normalize RMS. This gives background surrogate n_t. It preserves the coarse measured spectral envelope but not transients, phase, or within-hour events. It must not be described as recovered raw audio.

For the field experiment we use the corresponding continuous 8-kHz SanctSound hydrophone recordings directly: 6 hours each from MB03_01, 02, and 03, and a continuous 24-hour MB03_04 test day (42 hours total). One-second Hann periodograms are integrated into 23 one-third-octave bands from 25 to 4000 Hz, averaged to 5-s features, and power-aggregated to 30 s, 1 min, or 10 min. No waveform phase is synthesized in this experiment.

The original injected signal s_t follows the controllable synthesis design developed in MINUS: harmonic partials plus band-limited noise, attack/release edges, and low-frequency amplitude modulation. The broad profile adds chirps, inharmonic tones, impulse trains, frequency-modulated tones, and band-limited noise, while expanding carrier, modulation, envelope, transient, and noise-mix ranges. Parameters, family, and source-to-background SNR are drawn deterministically for every sample. The mixture is $x_t = n_t + s_t$, and clipping is prevented by common rescaling.

3 Method

Let $S_{t-H:t-1} \in \mathbb{R}^{H\times F}$ be normalized historical TOL. A GRU produces

$$z_t = \mathrm{LN}(\mathrm{GRU}(S_{t-H:t-1})), \quad (1)$$

$$\widehat{S}_t = g_\theta(z_t). \quad (2)$$

Only observations ending at $t-1$ enter the encoder. All three injection ablations use $c_t = \widehat{S}_t$; the baseline omits the forecaster.

BS-RoFormer converts mixture STFT bins into band tokens $X \in \mathbb{R}^{B\times T\times K\times D}$. Let $P_k(c_t)$ be a learned band-specific projection and $E(c_t)$ a sequence of embedded forecast bins. We compare three residual injection operators at the same point before the time/frequency Transformer:

$$\text{Concat: } \widetilde{X}_k = X_k + f([X_k; P_k(c_t)]), \quad (3)$$

$$\text{FiLM: } \widetilde{X}_k = (1+\tanh\gamma_k(c_t))\odot X_k + \beta_k(c_t), \quad (4)$$

$$\text{Cross-attn: } \widetilde{X} = X + \mathrm{Attn}(X, E(c_t), E(c_t)). \quad (5)$$

The residual branches are initialized to zero, so every run begins from the same audio mapping. Concatenation doubles the local feature width before a fusion MLP; FiLM directly changes band-token gain and offset; cross-attention learns content-dependent alignment to forecast-frequency tokens.

The end-to-end objective is

$$\mathcal{L} = \mathcal{L}_{\text{wave}} + \tfrac{1}{2}\sum_r \mathcal{L}_{\text{complex STFT},r} + \lambda\|\widehat{S}_t - S_t\|_1, \quad (6)$$

with $\lambda = 0.2$ for conditioned variants. Gradients from both objectives reach the forecast encoder; this is joint training, not frozen feature export.

4 Experimental protocol

All variants use identical samples, synthesis seeds, optimizer schedule, model width, matched separator-core initialization, and checkpoint rule. The checkpoint with maximum validation SI-SDR improvement (SI-SDRi) is evaluated once on MB03_04. SI-SDR follows the scale-invariant projection form; improvement subtracts mixture SI-SDR. We also report residual-background SI-SDR, forecast MAE in dB, inference real-time factor (RTF), parameter count, and the fraction of paired test examples where conditioning beats the baseline. Metric uncertainty is a seeded paired bootstrap over test examples. BSS evaluation background is discussed by Vincent et al. [6].

For distribution shift, waveform power is pooled into $K = 64$ log-frequency bins and normalized to p. Let q be the mean distribution of 512 draws from the original narrow harmonic generator. We report the directional $D_{\mathrm{KL}}(p\|q)$ and use the finite symmetric quantity

$$D_{\mathrm{JS}}(p,q) = \tfrac{1}{2}D_{\mathrm{KL}}(p\|m) + \tfrac{1}{2}D_{\mathrm{KL}}(q\|m), \quad m = (p+q)/2, \quad (7)$$

as the primary shift metric [3]. Because distance from the old generator need not measure source complexity, we also report normalized spectral entropy $H(p)/\log_2 K$ and source–ocean histogram intersection $\sum_k \min(p_k, n_k)$.

Table 1: Matched 2,000-update broad-source ablation.

Injection	Val. SI-SDRi (dB)	Test SI-SDRi (dB)	Paired Δ_0 (dB)	Params (k)
None	4.96	5.00	0	231
Concatenation	5.17	5.30	+0.15	238
FiLM	**5.35**	**5.39**	**+0.34**	255
Cross-attention	4.99	5.11	+0.01	220

4.1 Conditioning ablation and selection

Each model uses 4,000 examples per training deployment, 2,000 updates, and SNR drawn uniformly from $[-12, 12]$ dB. We select only by MB03_03 validation SI-SDRi: FiLM obtains 5.35 dB, versus 5.17 for concatenation and 4.99 for cross-attention. On the ordinary 256-example broad MB03_04 test, their scores are 5.39, 5.30, and 5.11 dB, respectively; the unconditioned model reaches 5.00 dB. Thus the held-out test confirms, rather than determines, the choice.

At fixed 0 dB, FiLM gains 0.34 dB over the same unconditioned outputs ($[0.21, 0.48]$), concatenation gains 0.15 dB ($[0.02, 0.27]$), and cross-attention gains 0.01 dB ($[-0.08, 0.10]$). FiLM is a good fit because a background spectrum naturally specifies which band features to attenuate, amplify, or shift. It preserves the audio evidence while applying a low-depth multiplicative/additive prior. Concatenation must relearn a compression of doubled features, while cross-attention must estimate an alignment from only a coarse hourly spectrum; its extra flexibility is not rewarded at this data scale. This explanation is mechanistic and consistent with the results, but is not a causal proof.

Table 2: Separation by injected-source strength (128 examples per level).

Input SNR (dB)	Cond. SI-SDR (dB)	Cond. SI-SDRi (dB)	Base SI-SDRi (dB)
-12	-4.63	7.30	7.32
-6	1.00	6.95	6.78
0	5.85	5.81	5.40
+6	9.55	3.51	2.94
+12	11.94	-0.10	-0.70

At weak and equal-strength inputs the selected FiLM model improves the mixture by at least 5.81 dB. At +12 dB, SI-SDRi is near zero because the mixture is already target-dominant, while absolute target SI-SDR remains 11.94 dB. This motivates separate improvement and absolute-quality gates rather than one threshold for all input strengths.

4.2 Broad-source distribution study

To isolate source shape from strength, the main diversity evaluation fixes SNR at 0 dB and uses 128 held-out

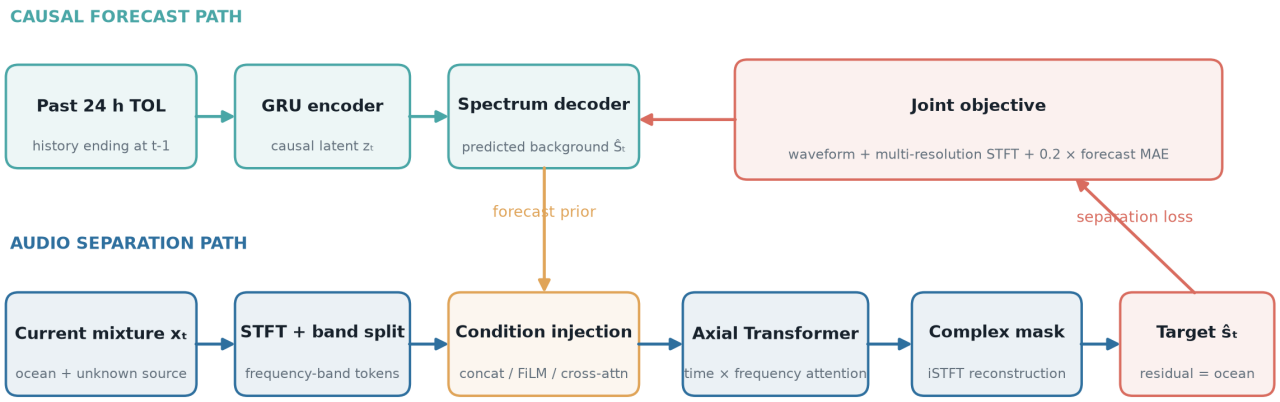


Figure 1: FC-BSR architecture. Historical TOL ending at $t-1$ is encoded into a forecast of the current ocean spectrum. Concatenation, FiLM, and cross-attention inject that prior before the axial Transformer. Separation and forecast losses train the system jointly; the unconditioned baseline omits the forecast path.

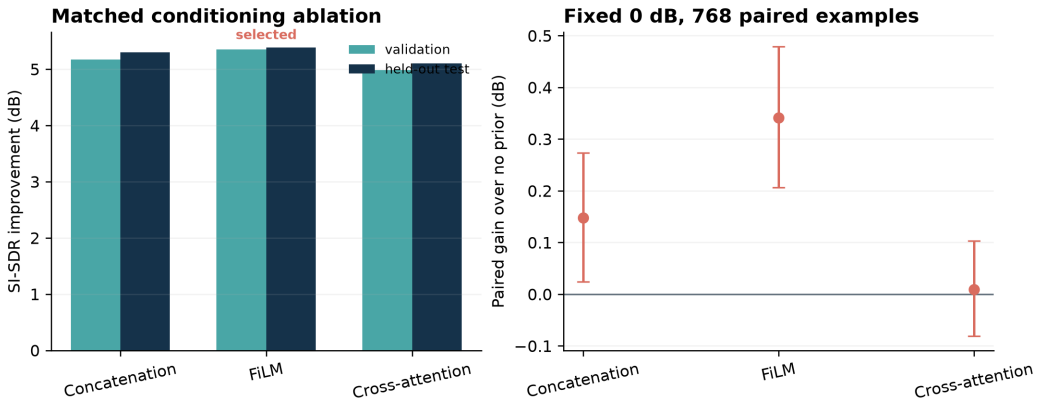


Figure 2: Matched injection ablation. Model selection uses MB03_03 validation only (left). On 768 fixed-0-dB paired MB03_04 examples (right), FiLM has the largest gain and its bootstrap interval excludes zero; cross-attention does not.

MB03_04 examples from each family. The resulting 768 examples are exactly paired in ocean waveform, source waveform, timestamp, and SNR.

Across all broad examples the baseline reaches 5.36 dB and FiLM reaches 5.70 dB, a paired gain of 0.34 dB. The gain–JS correlation is -0.085; the gain falls from 0.54 dB in the lowest JS quintile to 0.00 dB in the highest. Thus "farther from the original generator" is not a sufficient account of when an ocean prior helps.

Spectral entropy is more diagnostic. The gain–entropy correlation is 0.145; the highest entropy quintile gains +0.80 dB ($[0.47, 1.17]$). By family, band-limited noise gains +0.54 dB ($[0.19, 0.92]$), impulse trains gain 0.47 dB ($[0.16, 0.79]$), and frequency-modulated targets gain 0.43 dB ($[0.08, 0.82]$). FiLM therefore uses the prior broadly, with its strongest benefit on diffuse sources; raw JS/KL distance is not the governing variable.

5 Sub-hour real-waveform study

The sub-hour experiment fixes history coverage at one hour, giving 720, 120, 60, or 6 causal history steps. For target window t, the model receives only $S_{t-H:t-1}$; the background target is the actual waveform N_t. A broad synthetic source is injected into a clip inside that window, $x_t = N_t + \alpha s_t$, so clean signal and noise references remain exact. Absolute clip times and injection seeds are shared across resolutions.

The GRU is first trained for up to 2,500 updates. Because short-window ocean spectra are highly persistent, its zero-residual initialization is the explicit step-0 persistence candidate; a learned checkpoint is retained only if it lowers MB03_03 forecast MAE. The selected predictor is then frozen while the FiLM separator trains for at least 3,000 and at most 5,000 updates, with the best MB03_03 SI-SDRi checkpoint evaluated once on MB03_04. Thus

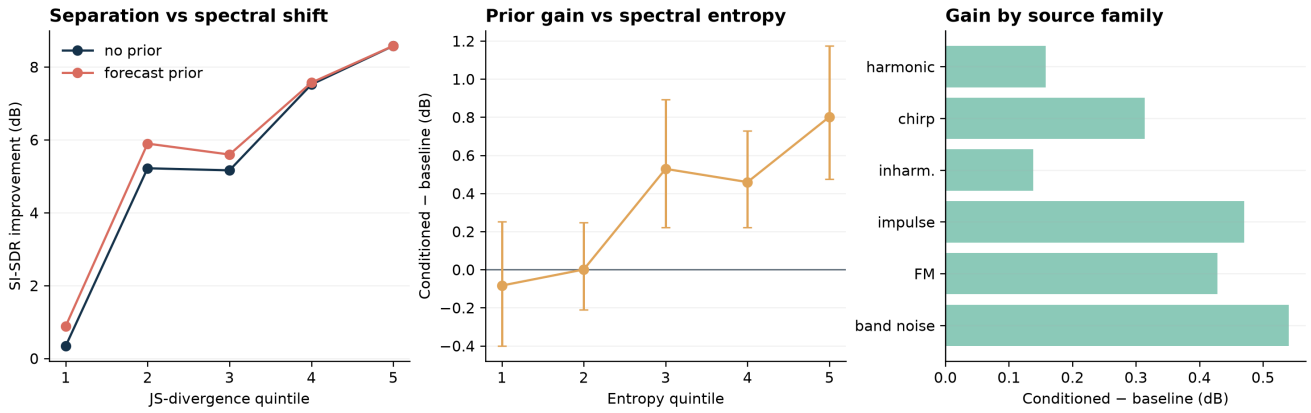


Figure 3: Broad-source results at 0 dB input SNR. Left: both models improve as JS-shift quintile increases, but the forecast prior does not close the gap. Middle: the prior becomes beneficial only in the most spectrally diffuse entropy quintile; bars show paired 95% bootstrap intervals. Right: family-wise FiLM gains are largest for band noise, impulses, and FM targets.

prediction quality cannot be silently degraded to optimize separation.

All four conditioned models pass the five-SNR engineering gate and run at about RTF 0.0024. The 5-s window has the highest absolute SI-SDRi and paired gain, but the four gain intervals overlap, so the experiment does not establish a significant ranking among resolutions. The important negative result is that more expressive prediction is not automatically useful: only 5 s improves over persistence on validation (by 1.4%), while 10-min training has only 60 distinct next-window examples and strongly overfits. Temporal background information is useful; with the present data, persistence supplies most of that information.

6 Reference-free evaluation on real events

Real event recordings do not provide isolated source and background stems, so SI-SDR is unavailable. We use the DCLDE 2013 North Atlantic right-whale corpus [1]: continuous 2-kHz mono WAV files are divided into 8-s clips, and recording days—not clips—define the split. Certain upcall annotations yield positives; negatives exclude every certain or possible call plus a 2-s margin. The balanced manifest contains 5,292 training, 2,696 validation, and 2,676 test clips. Only clean training/validation backgrounds receive synthetic, time-localized upcalls for supervised separator training. The 1,338 real positive test clips never participate in checkpoint selection.

Following learned referenceless separation assessment [2] and recent semantic reference-free metrics [7], we train a separate event classifier, calibrate its temperature on validation, and freeze it. Test AUROC is 0.9704, average precision 0.9738, and F1 0.9196. For event estimate $\hat{e}$ we enforce exact mixture consistency, $\hat{n} = x - \hat{e}$, then intervene with $y_g = \hat{n} + 10^{g/20}\hat{e}$ for $g \in \{-60, -30, -24, -18, -12, -6, 0\}$ dB. A credible output should reduce event probability in $\hat{n}$, retain event evidence in $\hat{e}$, recover that evidence monotonically as g rises, and minimally change samples outside annotated intervals. Identity ($\hat{e} = 0$) and all-to-event ($\hat{e} = x$) controls guard against gaming any one score.

The separator lowers event probability from 0.856 to 0.523 while its event stem scores 0.904; 98.6% of adjacent restoration steps are nondecreasing. Exact reconstruction error is 1.36×10^{-8}. The controls show why the semantic score is insufficient alone: all-to-event removes the classifier label but changes all off-event audio, whereas identity preserves audio but removes nothing. The separator changes off-event audio by 9.4%, but 59.7% of its stem energy lies outside annotated intervals. Some of this reflects incomplete event boxes and reverberation, but the value is too high for a deployment claim; this is evidence of semantic removal, not clean real-source recovery.

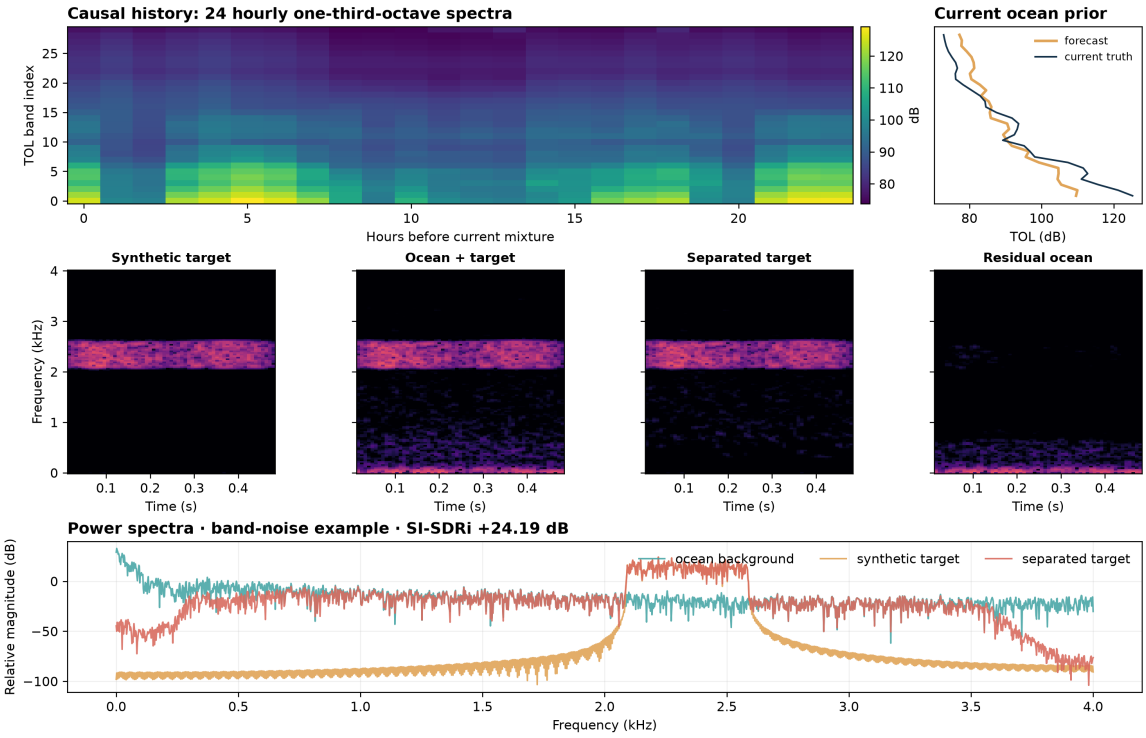


Figure 4: Representative high-entropy band-noise example. The causal TOL history and current forecast are shown above; source, mixture, separated target, and residual-ocean spectrograms are shown in the middle; spectra are overlaid below. The example illustrates how the selected FiLM prior removes low-frequency ocean energy while retaining a diffuse band target.

Table 3: Sub-hour prediction and separation on continuous real ocean audio. Intervals are paired 95% bootstrap intervals over 1,024 held-out clips.

Window	History steps	Selected forecast	Val. MAE (dB)	Cond. SI-SDRi (dB)	Base SI-SDRi (dB)	Paired gain (dB)	95% interval (dB)
5 s	720	GRU, step 1600	1.366	**29.753**	28.527	**+1.226**	[1.082, 1.381]
30 s	120	persistence	0.782	29.630	28.534	+1.095	[0.935, 1.265]
1 min	60	persistence	0.687	29.661	28.533	+1.128	[0.971, 1.298]
10 min	6	persistence	1.371	29.711	28.535	+1.176	[1.036, 1.326]

Table 4: Frozen-classifier and partial-GT diagnostics on 1,338 real calls.

Estimate	$p(x)$	$p(\hat{n})$	$p(\hat{e})$	Mono.	Off-event error
Separator	.856	.523	.904	.986	.094
Identity	.856	.856	.509	1.000	.000
All-to-event	.856	.509	.856	.793	1.000

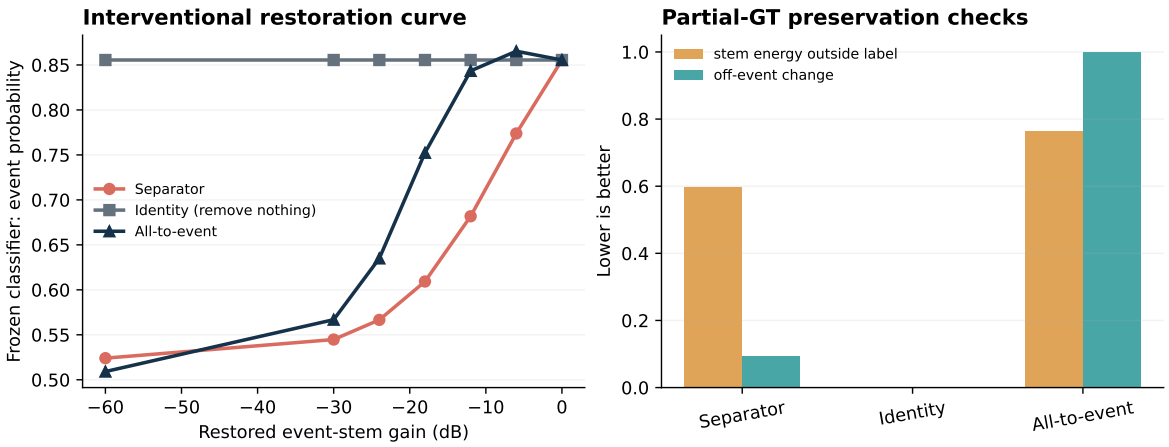


Figure 5: Real-event evaluation without separated ground truth. Left: restoring the separator event stem produces a nearly monotone increase in frozen classifier probability; identity is flat and all-to-event is non-monotone at high gain. Right: partial-GT preservation rejects the all-to-event degeneracy and quantifies the separator's remaining off-event leakage.

7 Usability criteria

The engineering gate requires overall SI-SDRi of at least 5 dB, at least 3 dB improvement for input SNR at or below 0 dB, at least 8 dB absolute target SI-SDR for stronger inputs, and frontend RTF below one. The conditioned model passes all four checks. Its batched RTF is 0.0040; the complete chunked frontend processes a 10-second WAV at RTF 0.043 on an RTX 3060. The scientific hypothesis is supported only if its positive paired SI-SDRi gain is reproduced on unseen, phase-aligned recordings. If forecast MAE improves while separation does not, the condition is either irrelevant or poorly injected. If oracle current spectra help but forecasts do not, forecasting is the bottleneck. These outcomes are explicitly falsifying, not post-hoc exceptions.

8 Local application

The repository includes the validation-selected one-megabyte FiLM checkpoint and a dependency-light local web application. A user supplies the last 24 hourly TOL rows and a WAV mixture. The application validates and normalizes history, predicts the current spectrum, separates overlapping waveform chunks, and exposes target and residual-ocean audio for playback and download. It also plots the predicted spectrum and reports duration, latency, and RTF. Requests stay on localhost; no recording or history is sent to a remote service.

9 Limitations

The real-waveform experiment removes the surrogate-background limitation, but the injected targets remain synthetic and all recordings come from one site. Only 12 training hours are available; at 10-min resolution this leaves 60 distinct next-window examples, which is inadequate for a flexible GRU. Multiple recording sites, seasons, training seeds, and real vessel, biological, and sonar targets are needed before deployment claims can be made.

The six source families are deliberately more diverse than the original generator, but JS divergence and entropy remain spectral summaries: neither captures semantic identity, long temporal organization, or recording-device effects. Family-wise findings should be confirmed with real vessels, animals, sonar, weather, and anthropogenic events before selecting a production model.

10 Conclusion

FC-BSR supplies a concrete, leakage-resistant route from soundscape forecasting to source separation and a usable local interface. FiLM is selected among three matched injection mechanisms. On continuous real backgrounds, all four sub-hour priors significantly improve separation; 5 s is best in absolute score with a +1.226 dB paired gain. Learned dynamics beat persistence only at 5 s, so current evidence supports temporal conditioning more strongly than a complex forecaster. On real annotated calls without isolated stems, the frozen-classifier intervention supports semantic removal but exposes substantial stem energy outside event intervals. Multi-site, multi-season recordings with stronger partial or full source references remain necessary for deployment.

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